

RetailSec AI

Next-Gen Autonomous Threat Intelligence for Digital Retail

MUHAMMAD ASLAM A

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The Problem: Retail is Under Attack

"Cybercriminals are Targeting Your Checkout"

- **Peak Traffic Risks:** malicious bots hide within Black Friday surges.
- **Loyalty Fraud:** Attackers drain points and gift cards from high-value accounts.
- **Silent Breaches:** Average detection time is **200+ days**. Retailers lose brand trust instantly.
- **Analyst Burnout:** Security teams drown in false positives during sales events.

Beyond Standard Attacks

1. **Gift Card Cracking:** Brute-forcing card numbers to steal balances.
2. **Inventory Hoarding:** Scalper bots buying up limited stock (PS5/Sneakers).
3. **Account Takeover (ATO):** Using stolen credentials to order goods.
4. **Price Scraping:** Competitor bots aggressively scraping pricing data.

Proposed Solution: RetailSec AI

An AI Analyst that Understands Commerce We build a specialized intelligence layer that distinguishes between a *loyal customer* and a *sophisticated bot*.

Context Aware Knows that 50 logins/min is normal for a sale, but abnormal for a single user.

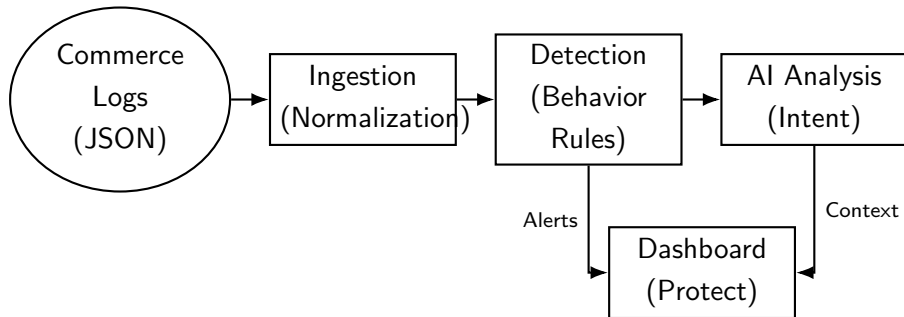
Generative Explanation Llama-3 explains: "*Blocking user for hoarding inventory during flash sale.*"

Instant Defense Blocks IP addresses at the edge before inventory is lost.

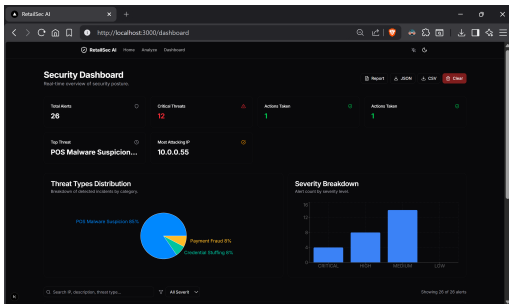
Powered by Modern Tech Stack

- **Frontend:** Next.js 14 (App Router) + Tailwind CSS v4.
- **Validation:** Zod for runtime type safety on log ingestion.
- **Visualization:** Recharts for interactive threat velocity graphs.
- **AI Core:** Groq SDK integration for sub-second LLM inference.
- **Storage:** LocalStorage persistence for session-based analysis.

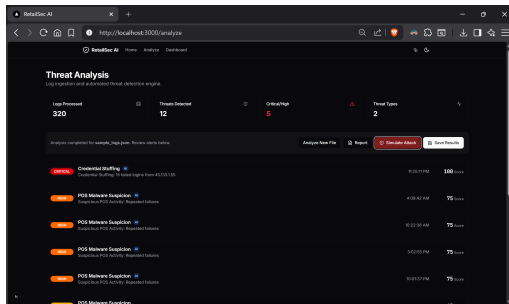
Process Flow



Live Dashboard Preview



- Left: Security Dashboard



- Right: Threat Analysis

Why RetailSec AI is Different

Competitive Edge

- **Retail-Specific Modeling:** Trained on commerce threats (Gift Cards, Inventory), not just generic WAF rules.
- **GenAI Explanations:** Llama-3 explains *why* a block happened, reducing analyst fatigue.
- **One-Click Response:** Instant remediation directly from the dashboard.
- **Lightweight:** Runs on Vercel Edge (Next.js), no heavy SIEM infrastructure required.

"Stopping a Loyalty Hack"

1. **Ingest:** Upload `checkout_logs.json`.
2. **Analyze:** System flags "High-Velocity Gift Card Checks".
3. **Explain:** **AI Agent** analyzes behavior:
"User checking 100+ gift cards with 0 purchases. Pattern matches 'Card Cracking'."
4. **Respond:** One-click "Block IP" prevents financial loss.

Impact & Benefits

- **Revenue Protection**

Stops inventory loss and chargebacks.

- **Customer Trust**

Ensures real humans get the products, not bots.

- **Operational Efficiency**

AI reduces triage time by 90%.

- **Cost Efficient**

Open-source AI (Llama-3) is cheaper than fraud losses.

Future Scope: Roadmap to Production

- **Live Streams:** Integrate Apache Kafka during Black Friday.
- **Auto-Remediation:** Webhooks to update Cloudflare WAF / AWS Shield.
- **Federated Learning:** Share fraud patterns with other retailers securely.
- **Voice Interface:** "Hey RetailSec, did we lose any inventory to bots today?"

Thank You!
Questions?